MUJEEB UR RAHMAN KHAN

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Portfolio | LeetCode | Linkedin | GitHub | Blog

Experience

Tata Consultancy Services (TCS) | *Software Engineer* Mumbai, India | Jan 2025 – Present

- Currently part of Engineering Team at National Stock Exchange, shaping robust digital platforms for India's financial ecosystem
- Hands-on exposure to **process management**, **queues**, **shared memory**, and **IPC mechanisms**, to understand and analyze low-latency system workflows.
- Developed **shell scripts** to **automate routine operations**, enhancing system efficiency and reliability.
- Performed **module-level testing and debugging** using GDB, log analysis, and runtime diagnostics.
- Led a **knowledge transfer sessions**, technical documentation, and mentoring of junior team members.

Projects

Low-Latency Logger | Link *C++, Threads, Atomics, FStream*

- Built a **high-performance asynchronous logging system** using a **ring buffer** and **background thread** to decouple logging from I/O
- Achieved **1M logs in 82ms** with **80ns enqueue latency per log** through **batching**
- Implemented **lock-free, non-blocking logging** for producer threads under heavy load
- Reduced **CPU usage** by preventing **busy-waiting** with **adaptive backoff** in the consumer thread
- Ensured **graceful shutdown** with **queue draining**, guaranteeing **zero log loss** on exit

End-to-End Chat Application *C++, TCP/IP, Socket Programming*

- Built a **real-time client-server chat application** using **C++ socket programming**, enabling **end-to-end communication** over **TCP**.
- Designed **multi-client message handling**, implementing **reliable data transfer**, connection lifecycle management, and **graceful disconnects**.
- Improved **performance and reliability** through efficient I/O handling and robust resource management (**socket cleanup**, timeouts, error handling).

QR Code Scanner | Link *Android, Kotlin, Room db, MVVM*

- Developed a modern Android app with supports for multiple QR formats, integrated scan history and Google scanner API
- **Published on Google Play Store with over 3000+ downloads** and consistently active userbase
- Implemented **MVVM architecture with Room Database** to store and manage scan history efficiently, ensuring lifecycle-aware, maintainable code

Education

P.E.S MODERN COLLEGE OF ENGINEERING Pune, India
Bachelors of Engineering IT, *CGPA: 8.65* July 2020 - July 2024

KENDRIYA VIDYALAYA NO.2 Pune, India
Class 12, *Percentage: 68%* April 2018 - May 2019

Skills

Programming Languages:	C/C++, Core Java, HTML/CSS
Libraries/Frameworks:	React, TailWind CSS
Tools / Platforms:	Android Studio, VS Code, Git, Github, Postman, Figma
Databases:	MySQL, Firebase

Honors & Awards

- 1st Rank at ASPIRE 2024 Hackathon
- 2nd Runner-Up at Hackathon, PES MCOE 2024
- Avishkar 2023 college level winner and participated in zonal level